Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

MAT 2143 – The Final Exam **Instructor: K. Zaynullin**

Last name: _____

First name: _____

Signature: _____

Student number:

Please, read the following instructions carefully:

- You have 3 hours to complete this exam. Read each question carefully. Where it is possible to check your work, do so.
- This is a closed book exam, and no notes of any kind are allowed. The use of calculators, cell phones, players, computers or any text storage or communication device is not permitted.
- Answer all questions in the space provided after each question. Each answer requires justification written legibly and logically: you must convince the marker that you know why your solution is correct. Use the backs of pages if necessary.

THIS SPACE IS RESERVED FOR THE MARKER:

Question	1	2	3	4	5	6	7	8
/out of	/3	/3	/3	/2	/2	/3	/3	/2

Question	9	10	11	12	13	14	15	Total
/out of	/2	/3	/2	/3	/2	/2	/2	/37

585, av. King-Edward
Ottawa (Ontario) K1N 6N5 Canada

585 King Edward Avenue
Ottawa, Ontario K1N 6N5 Canada

(613) 562-5864 • Téléc./Fax (613) 562-5776
Courriel/Email: uomaths@science.uottawa.ca

1. Determine whether the relation R is an equivalence relation on the set A, where

$$A = \mathbb{R} \times \mathbb{R} \quad \text{and} \quad (x_1, y_1)R(x_2, y_2) \text{ means that } x_1^2 + x_2^2 = y_1^2 + y_2^2.$$

Answer: (1)

Justification: (2)

2. Find $x \in \mathbb{Z}$ such that

$$x \equiv 3 \ (mod\ 9), \quad x \equiv 8 \ (mod\ 10), \quad \text{and } x \equiv 2 \ (mod\ 7).$$

Answer: $x =$ (1)

Solution: (2)

3. Find all integers $k \geq 2$ such that $k^2 - 5k \equiv 0 \mod 15$.

Answer: (1)

Solution: (2)

4. Let $\sigma = (1\ 2\ 3\ 4\ )(5\ 6\ 7)$ be a permutation in S_7.

(a) Factor σ^{-1} into a product of disjoint cycles

Answer: (1)

(b) Factor σ into a product of transpositions

Answer: (1)

5. Show that the transposition $(2\ 3) \in S_6$ can not be written as a product of cycles of length 3.

Solution: (2)

6. Let G be a group and let $a, b \in G$. We define a subset $H = \{g \in G \mid bga = agb\}$. Show that if $ab = ba$, then H is a subgroup of G.

Solution: (3)

7. Let H and K be subgroups of a group G. Prove that $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.

Solution: (3)

8. Let $G = \langle g \rangle$ be an infinite cyclic group and let $A = \langle g^a \rangle$ and $B = \langle g^b \rangle$ be cyclic subgroups of G (here $a, b \in \mathbb{Z}$). Prove that $A \subseteq B$ if and only if $a = qb$ for some $q \in \mathbb{Z}$.

Solution: (2)

9. Let $G = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \mid a, b \in \mathbb{R},\ a \text{ and } b \text{ are not both zero} \right\}$.

Show that G is isomorphic to the group $(\mathbb{C}^\times,\ \cdot\ )$ of non-zero complex numbers with respect to multiplication.

Solution: (2)

10. Let G be a group of order pq, where p and q are prime numbers. Let H be a non-trivial subgroup of G such that $H \neq G$. Show that H is a cyclic subgroup.

Solution: (3)

11. Let $N \triangleleft G$ and $K \triangleleft G$ be normal subgroups in G. Show that $N \cap K \triangleleft G$.

Solution: (2)

12. If $K \triangleleft G$ has prime index p (that is $[G : K] = p$). Show that $G = K \cup Ka \cup \ldots \cup Ka^{p-1}$ is a disjoint union for some $a \in G$.

Solution: (3)

13. Let $f\colon G \to G_1$ be a group homomorphism and let $a \in G$. We denote $K = \ker f$.

Show that $Ka = \{g \in G \mid f(g) = f(a)\}$.

Solution: (2)

14. Find the characteristic of the ring $M_2(\mathbb{Z}/n\mathbb{Z})$ of 2×2-matrices with entries in $\mathbb{Z}/n\mathbb{Z}$, $n \geq 2$.

Answer: (1)

Justification: (1)

15. Let R be a ring. Show that if $n \geq 2$, then the ring of matrices $M_n(R)$ is not a domain.

Solution: (2)